Joseph M. Luken

15819 S 7th Dr Phoenix, AZ 85045| 480-343-1120 | joe.luken@gmail.com | https://github.com/loejuken

Education

Georgia Institute of Technology | Atlanta, GA *August 2026 – Present*
Bachelor of Science in Electrical Engineering Expected Graduation: May 2029

Desert Vista High School | Phoenix, Arizona *August 2022 – May 2026*
GPA 4.895/5 (Weighted)

Skills

PCB/EDA: KiCad, Schematic Capture, PCB Layout, PCB Design, ERC/DRC
Hardware: Microcontrollers, Electronics Prototyping, Hardware Debugging, Component Selection, ATmega32U4, USB/USB-C, Soldering
CAD: Fusion 360, Parametric 3D Modeling, Mesh 3D Modeling, 3D Printing
Programming: Python, MATLAB, Java
Development: Git, GitHub

Projects

Custom TrackPoint Mechanical Keyboard | Personal Project *Summer 2026*
Sole Developer — Hardware, Firmware & Embedded Systems

Designed custom 63-key mechanical keyboard incorporating a TrackPoint pointing device, custom PCB, and embedded microcontroller with peripheral circuitry.

- Developed a 2-layer, 90+ component PCB in KiCad around an ATmega32U4, integrating USB-C, switch matrix, TrackPoint, and supporting circuitry.
- Programmed and configured Quantum Mechanical Keyboard (QMK) firmware to implement keyboard and TrackPoint functionality through the custom hardware and USART interface for tracking communication.
- Designed and iterated a 3D-printed keyboard plate and enclosure in Fusion 360, refining friction-fit tolerances and ergonomics through 3 physical prototypes.
- Manufactured and assembled the PCB and enclosure, performing soldering, hardware debugging, and firmware testing to produce a functional keyboard and tracking device in one piece of hardware.

Leadership or Activities

Gold Medal Swim Club | Competitive Swimmer & Team Captain *August 2020 – August 2026*

- 3-time individual State Champion, multi-time finalist, and Sectionals & Futures qualifier through year-round competitive training exceeding 25 hours per week.
- Led and mentored 10 younger athletes as team captain for 2 years, providing technique coaching and guidance while cultivating a competitive and supportive team culture.